EE3111: Statistics for Electrical Engineers

Lab 6: Linear Statistical Models

Multiple Linear Regression & Polynomial Regression

Jan–May 2026

Due date: April 8, 2026

Dataset: Concrete Compressive Strength (UCI)

We will use the **Concrete Compressive Strength** dataset: 1030 concrete mixes with 8 input variables and compressive strength as the target.

- Dataset page (description + variables): https://archive.ics.uci.edu/dataset/165/concrete+compressive+strength
- Direct download (ZIP): https://archive.ics.uci.edu/static/public/165/concrete%2Bcompressive%2Bstrength.zip

Variables (typical column names)

- Inputs: Cement, Slag, Fly Ash, Water, Superplasticizer, Coarse Aggregate, Fine Aggregate, Age (days)
- Output: Concrete compressive strength (MPa)

Part A: Data Setup & EDA

1. Load the dataset and check for missing values.
2. Create **two** engineered features (choose any two):
   - Water–cement ratio: $w/c = \frac{\text{water}}{\text{cement}}$
   - Binder: $\text{binder} = \text{cement} + \text{slag} + \text{fly_ash}$
   - Fine/Coarse ratio: $\frac{\text{fine_agg}}{\text{coarse_agg}}$
3. Make **2–3 plots** (with 2–3 sentences each):
   - Strength vs Age (scatter)
   - Strength vs w/c (scatter)
   - Correlation heatmap (or pairplot of key predictors)
4. Write 3 bullet points summarizing: strongest predictors, visible nonlinearity, and any outliers/strange points.

Part B: Multiple Linear Regression (MLR)

B1. Train/test split

Split the data into training and test sets (e.g., 80/20). Set a random seed for reproducibility.

B2. Fit a baseline MLR model

Fit an OLS model predicting `strength` using the **8 original predictors**. Optionally, fit a second model that also includes your engineered features, but keep the baseline for comparison.

Required outputs:

- Coefficient table (estimate, standard error, p-value).
- R^2 and adjusted R^2 on training set. Read about R^2.
- Test-set RMSE.
- Residual plot on test set (residual vs fitted or residual vs Age).

B3. "Is predictor X still useful after controlling for others?"

Choose **one** predictor (suggestions: `superplasticizer` or `fly_ash`). Test:

$$H_0 : \beta_X = 0 \qquad \text{vs.} \qquad H_1 : \beta_X \neq 0$$

Report:

- the p-value and a 95% confidence interval for β_X,
- an interpretation in words with units and direction.

Part C: Polynomial Regression for Nonlinearity in Age

Nonlinearity is common in materials: strength increases with curing time but not always linearly.

C1. Fit three nested models (same predictors, different Age terms)

Keep the same predictors as Part B, but change only how Age enters:

- Model 1: Age
- Model 2: Age + Age^2
- Model 3: Age + Age^2 + Age^3

Required outputs:

- A small table with: model, train adj-R^2, test RMSE.

C2. Interpret

- Which polynomial degree is best by test RMSE?
- Any signs of overfitting (e.g., Model 3 improves train fit but worsens test RMSE)?

Part D: Engineering Decision (Prediction with Uncertainty)

A client requires **strength $\geq$ 40 MPa at 28 days**. Using your **best** model:

1. Select any 5 rows (mix designs) from the dataset as "candidates".
2. For each candidate, set `age = 28` and predict strength.
3. Compute a **95% prediction interval** for each prediction (statsmodels has built-in options; bootstrap is also allowed).
4. Decide which mixes "Pass" the requirement *considering uncertainty*. Present a small table: candidate ID, predicted strength, PI lower/upper, Pass/Fail.
5. Write a short note recommending 1–2 mixes and explaining why.

Helpful Links (function signatures and docs)

- OLS (formula interface): https://www.statsmodels.org/stable/generated/statsmodels.formula.api.ols.html
- ANOVA / nested-model comparison: https://www.statsmodels.org/stable/anova.html
- Prediction intervals (statsmodels): https://www.statsmodels.org/stable/generated/statsmodels.regression.linear_model.RegressionResults.get_prediction.html
- Polynomial features (scikit-learn, optional): https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.PolynomialFeatures.html